Riemannian Geometry of Two-Agent Interaction Dynamics: A Unified Framework from Induced Metrics to Geometric Field Equations

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Abstract

We present a unified, self-contained geometric theory of two-agent interaction dynamics in three successive stages. *Stage I* constructs a time-dependent Riemannian metric $G(t) = I + \gamma(t)^\top \gamma(t)$ on the interaction manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{I_S}$ by imposing the linear constraint $I_P = \gamma(t) I_S$ on the passive interaction variable; all curvature arises from temporal deformation of G, in analogy with cosmological (FLRW) metrics. *Stage II* extends the framework to a state-dependent coupling $\gamma = \gamma(t, I_S)$, generating non-trivial spatial curvature through the interaction Jacobian $J = \gamma + (\nabla_{I_S}\gamma) I_S$, and decomposes the Ricci tensor into independent temporal and spatial contributions. *Stage III* elevates the active interaction vector I_S to a tangent vector field on a Riemannian manifold, identifies the interaction flow matrix $M(t)$ with the Levi-Civita connection, and proves—via the Jacobi equation—that the emotion quantity tensor $E = \dot{M} + M^2$ equals the geodesic deviation operator. Building on this geometric foundation, we prove under four explicit axioms (locality, diffeomorphism covariance, metric dynamics, and second-order field equations) that the unique admissible action functional on a three-dimensional interaction manifold is the Einstein–Hilbert action, whose variation yields the interaction field equation

$$R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu},$$

structurally identical to the Einstein field equations. The stress-energy tensor $T_{\mu\nu}$ is constructed explicitly from the gradient of the interaction scalar field $\varphi = \|I_S\|_g$. All prior linear results are recovered as the flat-space limit $R_{\mu\nu} = 0$. We discuss the dimensional subtlety for the two-agent dyadic case ($n = 2$), where the Einstein tensor vanishes identically and the correct field equation is the Liouville equation on the conformal factor of the metric.

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1 Introduction

Mathematical descriptions of interacting dynamical systems typically rely on linear equations defined over Euclidean state spaces. When interaction variables mutually constrain one another, however, the underlying state space acquires a non-trivial geometric structure: curvature then provides a coordinate-invariant measure of the structural deformation induced by temporal or spatial variation in the coupling.

Three prior papers by Pinmook [10–12] established a progressive algebraic hierarchy for psychophysical interaction:

$$\begin{aligned} &\text{(Discrete)} \quad I_S(t+1) = \gamma\, I_S(t), \\ &\text{(Continuous)} \quad \dot{I}_S(t) = M(t)\, I_S(t), \\ &\text{(Emotional)} \quad \ddot{I}_S(t) = E(t)\, I_S(t), \quad E(t) := \dot{M}(t) + M(t)^2. \end{aligned}$$

Despite this progression, the objects $I_S, I_P \in \mathbb{R}^n$ and the matrix $M(t)$ remain coordinate-dependent linear-algebraic entities. Three structural limitations follow: (i) the equations change form under nonlinear re-parameterisation of the interaction space; (ii) the Weber–Fechner logarithmic encoding naturally induces a non-Euclidean (hyperbolic) geometry on stimulus space [1] that is discarded by the flat $\mathbb{R}^n$ model; and (iii) no field equation is given for the source of $E(t)$.

The present paper addresses all three limitations simultaneously. We develop the geometry in three stages of increasing generality, each stage being a strict extension of the previous one. The main contributions are:

1. **Induced metric (Stage I).** A Riemannian metric is constructed from the linear constraint $I_P = \gamma(t) I_S$ via an extended Euclidean embedding; the interaction flow tensor $K^i{}_j = \frac{1}{2} G^{ik} \dot{G}_{kj}$ and the emotion quantity tensor $E_{ij} = \dot{K}_{ij} + K_{ik} K^k{}_j$ emerge as natural geometric objects governing all Ricci curvature components.

2. **State-dependent metric (Stage II).** Extending γ to $\gamma(t, I_S)$ introduces non-trivial spatial curvature; the Ricci tensor decomposes cleanly into temporal and spatial contributions.

3. **Emotion–curvature theorem (Stage III).** We prove that $E = \dot{M} + M^2$ is precisely the geodesic deviation operator on the interaction manifold, providing a coordinate-free identification of emotion as Riemannian curvature.

4. **Uniqueness of the interaction action.** Under four explicit axioms we prove that the Einstein–Hilbert action is the *unique* admissible action on the three-dimensional interaction manifold.

5. **Explicit stress-energy tensor.** The source tensor $T_{\mu\nu}$ is constructed from the canonical scalar-field action for $\varphi = \|I_S\|_g$, closing the field equations.

Organisation

Sections 2–3 develop Stage I. Sections 4–5 develop Stage II. Section 6 develops Stage III: the manifold lift, the emotion–curvature theorem, and the emotional stress-energy tensor. Section 7 contains the uniqueness theorem for the action and the derivation of the field equations. Section 8 presents special solutions with their interpretations. Section 9 discusses limitations, dimensional subtleties, and future directions.

2 Stage I: Time-Dependent Interaction Metric

2.1 Interaction Variables and Linear Constraint

Let two interaction vectors

$$I_S(t) \in \mathbb{R}^2, \qquad I_P(t) \in \mathbb{R}^2$$

represent the *active* and *passive* interaction components, respectively. We impose the *linear interaction constraint*

$$I_P(t) = \gamma(t)\, I_S(t), \tag{1}$$

where $\gamma(t) \in \mathbb{R}^{2\times 2}$ is a differentiable matrix-valued function of time. Under Weber–Fechner encoding [10], the matrix $\gamma_{ij} = k_{P,i}\alpha_i/k_{S,j}$ encodes the sensitivity ratio between agents, and (1) is exact in the linear regime.

2.2 Extended Euclidean Embedding

Consider the extended coordinate space

$$(t,\, I_P^1,\, I_P^2,\, I_S^1,\, I_S^2) \in \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2,$$

equipped with the standard Euclidean line element

$$\mathrm{d}s^2 = \mathrm{d}t^2 + \mathrm{d}I_P^\top \mathrm{d}I_P + \mathrm{d}I_S^\top \mathrm{d}I_S.$$

2.3 Induced Interaction Metric

Imposing (1) at fixed t gives $\mathrm{d}I_P = \gamma(t)\,\mathrm{d}I_S$. Substituting into the Euclidean line element yields

$$\mathrm{d}s^2 = \mathrm{d}t^2 + \mathrm{d}I_S^\top\big(I + \gamma(t)^\top \gamma(t)\big)\mathrm{d}I_S.$$

Definition 2.1 (Interaction Metric, Stage I)**.** The *spatial interaction metric* is

$$G(t) = I + \gamma(t)^\top \gamma(t). \tag{2}$$

The full metric on $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{I_S}$ is

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t) \end{pmatrix},$$

where $a > 0$ is a constant temporal scale factor.

By construction, $G(t)$ is symmetric and positive definite, and $\partial_k G_{ij} = 0$ (no spatial dependence in Stage I).

3 Stage I: Curvature from Temporal Deformation

Definition 3.1 (Interaction Flow Tensor)**.** The *interaction flow tensor* is

$$K^i{}_j := \tfrac{1}{2}\, G^{ik}\dot{G}_{kj}, \tag{3}$$

where $\dot{G}_{kj} = \mathrm{d}G_{kj}/\mathrm{d}t$. It measures the rate of temporal deformation of the interaction metric.

Theorem 3.2 (Non-vanishing Christoffel Symbols, Stage I)**.** *For the block-diagonal metric with $\partial_k G_{ij} = 0$, the only non-vanishing Christoffel symbols are*

$$\Gamma^0{}_{ij} = -\frac{1}{2a^2}\dot{G}_{ij}, \qquad \Gamma^i{}_{0j} = \Gamma^i{}_{j0} = K^i{}_j.$$

Proof. Apply the standard formula $\Gamma^\lambda{}_{\mu\nu} = \frac{1}{2}g^{\lambda\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$ to the block-diagonal metric with $g_{0i} = 0$, $g_{00} = a^2$ constant, and $\partial_k G_{ij} = 0$. The only surviving terms involve $\partial_t G_{ij}$. □

Definition 3.3 (Emotion Quantity Tensor)**.** The *emotion quantity tensor* is

$$E_{ij} := \dot{K}_{ij} + K_{ik}K^k{}_j. \tag{4}$$

Theorem 3.4 (Ricci Tensor, Stage I)**.** *The non-vanishing components of the Ricci tensor are*

$$R_{00} = -\operatorname{Tr}(\dot{K}) - \operatorname{Tr}(K^2), \tag{5}$$

$$R_{ij} = \dot{K}_{ij} + K_{ik}K^k{}_j + K_{ij}\operatorname{Tr}(K) = E_{ij} + K_{ij}\operatorname{Tr}(K). \tag{6}$$

Proof. Temporal component. Since $\Gamma^\lambda{}_{00} = 0$, the Ricci formula $R_{00} = \partial_\lambda\Gamma^\lambda{}_{00} - \partial_0\Gamma^\lambda{}_{0\lambda} + \Gamma^\lambda{}_{\lambda\sigma}\Gamma^\sigma{}_{00} - \Gamma^\lambda{}_{0\sigma}\Gamma^\sigma{}_{0\lambda}$ reduces to $R_{00} = -\partial_t\Gamma^i{}_{0i} - \Gamma^i{}_{0k}\Gamma^k{}_{0i} = -\operatorname{Tr}(\dot{K}) - \operatorname{Tr}(K^2)$, establishing (5).

Spatial components. Since $\partial_k G_{ij} = 0$, all purely spatial Christoffel symbols vanish. Computing each term in the Ricci formula using only $\Gamma^0{}_{ij}$ and $\Gamma^i{}_{0j}$ yields three surviving contributions: $-\partial_t(\Gamma^i{}_{j0}) = -\dot{K}^i{}_j$ from the second term, $\Gamma^i{}_{0k}\Gamma^k{}_{j0} = K^i{}_k K^k{}_j$ and $\Gamma^k{}_{0k}\Gamma^i{}_{j0} = \operatorname{Tr}(K)K^i{}_j$ from the cubic contractions, giving (6). □

Remark 3.5. In Stage I all curvature arises from temporal deformation of G. The metric is structurally analogous to a Friedmann–Lemaître–Robertson–Walker (FLRW) cosmological metric with $G_{ij}(t)$ playing the role of the scale factor matrix.

4 Stage II: State-Dependent Interaction Metric

4.1 Motivation for State Dependence

If $\gamma = \gamma(t)$ only, then $\partial_k G_{ij} = 0$ and the interaction manifold is spatially flat. Many realistic interaction systems require the coupling to depend on the current state:

- *Nonlinear feedback*: the coupling strength varies with interaction intensity (e.g., emotional escalation or saturation).
- *Biological saturation*: response functions saturate at high stimulation.
- *Adaptive feedback control*: systems adjust sensitivity based on current state.

4.2 Extended Interaction Constraint and Jacobian

With $\gamma = \gamma(t, I_S)$, the differential of the constraint $I_P = \gamma I_S$ is

$$\mathrm{d}I_P = \gamma\, \mathrm{d}I_S + (\partial_t \gamma) I_S\, \mathrm{d}t + (\partial_{I_S}\gamma)(\mathrm{d}I_S)\, I_S.$$

At fixed time ($\mathrm{d}t = 0$), define the *interaction Jacobian*

$$J(t, I_S) := \gamma(t, I_S) + (\nabla_{I_S}\gamma(t, I_S))\, I_S, \qquad \big((\nabla_{I_S}\gamma)\big)^{ij}{}_k := \frac{\partial \gamma^{ij}}{\partial I_S^k}\, I_S^k. \tag{7}$$

4.3 State-Dependent Metric

Definition 4.1 (Interaction Metric, Stage II)**.** The *state-dependent spatial interaction metric* is

$$G(t, I_S) = I + J(t, I_S)^\top J(t, I_S). \tag{8}$$

The full manifold metric retains the block-diagonal form

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t, I_S) \end{pmatrix}.$$

$G(t, I_S)$ is symmetric, positive definite, and depends on both t and I_S. Since $\partial_k G_{ij} \neq 0$ whenever $\nabla_{I_S}\gamma \neq 0$, the spatial geometry is generically non-flat.

5 Stage II: Connection and Curvature with State Dependence

Theorem 5.1 (Levi-Civita Connection, Stage II)**.** *For the state-dependent metric, the non-vanishing Christoffel symbols are*

$$\begin{aligned}
\Gamma^0{}_{ij} &= -\frac{1}{2a^2}\, \partial_t G_{ij}, \\
\Gamma^i{}_{0j} &= K^i{}_j = \tfrac{1}{2}\, G^{ik}\partial_t G_{kj}, \\
\Gamma^i{}_{jk} &= \tfrac{1}{2}\, G^{il}(\partial_j G_{kl} + \partial_k G_{jl} - \partial_l G_{jk})\,.
\end{aligned}$$

The purely spatial symbols $\Gamma^i{}_{jk}$ are non-zero whenever $\nabla_{I_S}\gamma \neq 0$.

Theorem 5.2 (Ricci Tensor Decomposition, Stage II)**.** *The Ricci tensor decomposes as*

$$R_{00} = -\operatorname{Tr}(\dot{K}) - \operatorname{Tr}(K^2), \tag{9}$$

$$R_{ij} = \dot{K}_{ij} + K_{ik}K^k{}_j + K_{ij}\operatorname{Tr}(K) + R^{\text{space}}_{ij}, \tag{10}$$

where R^{space}_{ij} is the Ricci tensor of G_{ij} computed with respect to its own Levi-Civita connection $\Gamma^i{}_{jk}$.

Proof. Since the metric is block-diagonal, the Ricci tensor splits by the general Gauss–Codazzi decomposition for warped products. The temporal components follow the identical calculation as Theorem 3.4 (the spatial Christoffel symbols contribute only to R_{ij}). The spatial piece R^{space}_{ij} is the intrinsic Ricci curvature of the submanifold $\{t\} \times \mathbb{R}^2_{I_S}$ equipped with $G_{ij}(t, \cdot)$. □

Remark 5.3. The decomposition (10) separates two geometrically independent sources of curvature:

- *Temporal curvature*: the $\dot{K}$ and K^2 terms arise from time-evolution of the interaction rule.
- *Spatial curvature*: R_{ij}^{space} arises from state-dependence of the coupling, i.e., from $\partial_k G_{ij} \neq 0$.

Theorem 5.4 (Non-trivial Curvature Condition)**.** *If $\nabla_{I_S}\gamma \neq 0$ in some open region of the interaction space, then R_{ij}^{space} is generically non-zero in that region.*

Proof. We have $\partial_k G_{ij} = (\partial_k J)^\top J + J^\top(\partial_k J)$. If $\nabla_{I_S}\gamma \neq 0$, then $\partial_k J \neq 0$, so $\partial_k G_{ij} \neq 0$. Non-vanishing spatial derivatives of the metric produce non-zero spatial Christoffel symbols $\Gamma^i{}_{jk}$, which in turn enter the curvature formula $R^i{}_{jkl} = \partial_k \Gamma^i{}_{lj} - \partial_l \Gamma^i{}_{kj} + \Gamma^i{}_{km}\Gamma^m{}_{lj} - \Gamma^i{}_{lm}\Gamma^m{}_{kj}$, giving non-zero R_{ij}^{space} generically. The special case $\nabla_{I_S}\gamma = 0$ recovers Stage I, where $R_{ij}^{\text{space}} = 0$. □

6 Stage III: Manifold Lift, Emotion–Curvature Theorem, and Scalar Field

6.1 The Interaction Configuration Manifold

The Stage I and II constructions embed the interaction space in a specific coordinate chart. Stage III places the framework in a coordinate-free setting.

Definition 6.1 (Interaction Configuration Manifold)**.** The *Interaction Configuration Manifold* is a pair $(\mathcal{M}, g)$ where $\mathcal{M}$ is a smooth connected n-dimensional manifold, each point $p \in \mathcal{M}$ represents a complete interaction configuration $(S_1, \ldots, S_n)$ of the agent system, and g is a smooth positive-definite metric tensor field on $\mathcal{M}$. A smooth curve $\sigma \colon \mathbb{R}^+ \to \mathcal{M}$ is an *interaction trajectory.*

Definition 6.2 (Active and Passive Interaction as Geometric Objects)**.** Let $\sigma \colon \mathbb{R}^+ \to \mathcal{M}$ be an interaction trajectory.

(i) The *active interaction field* is the velocity vector field $X(t) := \dot{\sigma}(t) \in T_{\sigma(t)}\mathcal{M}$, corresponding to $I_S(t)$ in local coordinates.

(ii) The *passive interaction form* is the metric dual $\omega(t) := g^\flat(X(t)) \in T^*_{\sigma(t)}\mathcal{M}$, with $\omega_\mu = g_{\mu\nu}X^\nu$, generalising $I_P = \gamma I_S$ via the identification $\gamma \leftrightarrow g^\flat$.

6.2 The Interaction Matrix as Connection Coefficients

Proposition 6.3 (Identification $M \leftrightarrow \Gamma$)**.** *In a local frame $\{e_i\}$ along $\sigma(t)$, the covariant derivative along σ acts as*

$$\frac{DX^i}{\mathrm{d}t} = \dot{X}^i + \Gamma^i{}_{jk}\dot{\sigma}^j X^k = \dot{X}^i + M^i{}_k X^k, \qquad M^i{}_k := \Gamma^i{}_{jk}\dot{\sigma}^j.$$

The interaction flow generator $M(t)$ from the continuous interaction equation $\dot{I}_S = MI_S$ is thus identified with the connection matrix along σ.

Corollary 6.4 (Covariant Interaction Equation)**.** *The equation $\dot{I}_S = MI_S$ lifts to the covariant equation*

$$\frac{DX}{\mathrm{d}t} = 0,$$

so $X = \dot{\sigma}$ is parallel-transported along σ and σ is a geodesic*:*

$$\ddot{\sigma}^\rho + \Gamma^\rho{}_{\mu\nu}\dot{\sigma}^\mu\dot{\sigma}^\nu = 0.$$

6.3 The Emotion–Curvature Theorem

Theorem 6.5 (Emotion as Geodesic Deviation)**.** *Let $\sigma(t)$ be a geodesic on $(\mathcal{M}, g)$ and let $J(t)$ be a Jacobi field along σ (an infinitesimal deviation between nearby geodesics). Then J satisfies the* Jacobi equation

$$\frac{D^2 J}{\mathrm{d}t^2} = R(\dot{\sigma}, J)\dot{\sigma}.$$

In a parallel orthonormal frame $\{e_i\}$ along σ, this takes the matrix form

$$\ddot{J}^i = \mathcal{E}^i{}_k\, J^k, \qquad \mathcal{E}^i{}_k \coloneqq R^i{}_{jkl}\, \dot{\sigma}^j \dot{\sigma}^l.$$

Under the identification $J \leftrightarrow I_S$, this is precisely the Emotion Equation $\ddot{I}_S = EI_S$, with

$$E(t) = \dot{M}(t) + M(t)^2 = \mathcal{E}(t). \tag{11}$$

Proof. Step 1. In a parallel frame along σ, set $M^i{}_k = \Gamma^i{}_{jk}\dot{\sigma}^j$. The covariant derivative along σ acts as $DX^i/\mathrm{d}t = \dot{X}^i + M^i{}_k X^k$. A direct computation gives

$$\frac{D^2 J^i}{\mathrm{d}t^2} = \ddot{J}^i + (\dot{M}^i{}_k + M^i{}_l M^l{}_k)\, J^k + M^i{}_k \dot{J}^k + M^i{}_k \dot{J}^k.$$

In the parallel frame the cross-terms $M\dot{J}$ cancel with the parallel-transport correction, yielding

$$\frac{D^2 J^i}{\mathrm{d}t^2} = \ddot{J}^i + (\dot{M} + M^2)^i{}_k J^k.$$

Step 2. By the definition of the Riemann curvature tensor, $D^2 J^i/\mathrm{d}t^2 = R^i{}_{jkl}\dot{\sigma}^j J^k \dot{\sigma}^l = \mathcal{E}^i{}_k J^k$.

Step 3. Comparing Steps 1 and 2: $\ddot{J}^i + (\dot{M} + M^2)^i{}_k J^k = \mathcal{E}^i{}_k J^k$. Setting $I_S \coloneqq J$ and noting that the $\ddot{J}^i$ term is absorbed into $\mathcal{E}$ by the Jacobi equation (that is, $\ddot{J}^i = (\mathcal{E} - \dot{M} - M^2)^i{}_k J^k$), we obtain $\dot{M} + M^2 = \mathcal{E}$. □

Remark 6.6. Theorem 6.5 resolves the causality ambiguity in [12]: the emotion tensor E is the Riemannian curvature of the interaction manifold, not an externally imposed forcing term. Positive sectional curvature forces nearby geodesics to converge (cohesive interaction); negative sectional curvature forces divergence (escalating conflict).

6.4 Sectional Curvature and Conflict

Definition 6.7 (Sectional Curvature)**.** For a two-dimensional plane $\Pi = \mathrm{span}\{u, v\} \subset T_p\mathcal{M}$:

$$K(\Pi) = \frac{R(u, v, u, v)}{g(u, u)g(v, v) - g(u, v)^2}.$$

Proposition 6.8 (Curvature and Conflict Parameter)**.** *For a two-agent system with flow matrix M and cross-influence conflict parameter $\mathcal{C} = -M_{AB}M_{BA}$, the sectional curvature of the interaction plane spanned by the two agent directions satisfies*

$$K = -\frac{\mathcal{C}}{\det g}$$

in normal coordinates at p. Consequently: $\mathcal{C} > 0$ (bidirectional cross-influence) gives $K < 0$ (hyperbolic, divergent geometry); $\mathcal{C} = 0$ gives $K = 0$ (flat, independent agents); $\mathcal{C} < 0$ (mutually inhibitory cross-influence) gives $K > 0$ (spherical, convergent geometry).

6.5 The Interaction Scalar Field and Stress-Energy Tensor

Definition 6.9 (Interaction Scalar Field)**.** Let $\varphi \colon \mathcal{M} \to \mathbb{R}$ be defined by $\varphi(p) = \|I_S(p)\|_g = \sqrt{g_{\mu\nu}X^\mu X^\nu}$, the *interaction scalar field* representing the magnitude of active interaction at each configuration point.

Definition 6.10 (Emotional Stress-Energy Tensor)**.** The *emotional stress-energy tensor* is the canonical stress-energy tensor of a massless scalar field on $(\mathcal{M}, g)$:

$$T_{\mu\nu} := \nabla_\mu\varphi\,\nabla_\nu\varphi - \tfrac{1}{2}\,g_{\mu\nu}\,g^{\alpha\beta}\nabla_\alpha\varphi\,\nabla_\beta\varphi, \tag{12}$$

derived from the action $S[\varphi, g] = \int_\mathcal{M} \frac{1}{2}|\mathrm{d}\varphi|_g^2\,\mathrm{dvol}_g$. It satisfies $\nabla^\mu T_{\mu\nu} = 0$ whenever $\Box_g\varphi = 0$.

Remark 6.11. High interaction gradients (rapid change of interaction magnitude) produce large $T_{\mu\nu}$, sourcing large curvature. Uniform φ (emotional equilibrium) gives $T_{\mu\nu} = 0$, hence flat geometry, recovering the prior linear framework.

7 Interaction Action Functional and Field Equations

7.1 Axioms of the Interaction Action

We impose four axioms on any governing action $S[g]$ for the interaction geometry.

Axiom 7.1 (Locality)**.** The interaction dynamics at each point depend only on the metric and its derivatives up to finite order at that point: $S[g] = \int_\mathcal{M} \mathcal{L}(g, \partial g, \partial^2 g, \ldots)\,\mathrm{dvol}_g$.

Axiom 7.2 (Diffeomorphism Covariance)**.** The action is invariant under smooth coordinate transformations: $S[g] = S[\phi^* g]$ for all $\phi \in \mathrm{Diff}(\mathcal{M})$.

Axiom 7.3 (Metric Dynamics)**.** The metric $g_{\mu\nu}$ is the sole dynamical geometric field; the Lagrangian depends only on tensors generated from g.

Axiom 7.4 (Second-Order Field Equations)**.** The Euler–Lagrange equations derived from $S[g]$ contain at most second derivatives of the metric.

7.2 Derivation of the Unique Action

Lemma 7.5 (Covariant Scalar Construction)**.** *Any diffeomorphism-invariant local Lagrangian satisfying Axioms 1–2 must be constructed from contractions of the Riemann tensor $R^{\rho}{}_{\sigma\mu\nu}$ and its covariant derivatives.*

Lemma 7.6 (Second-Order Constraint)**.** *If the Lagrangian contains ∇R or higher covariant derivatives of curvature, the resulting field equations contain derivatives of order ≥ 3 in the metric, violating Axiom 4.*

Proof. Since $R \sim \partial^2 g$, any $\nabla R \sim \partial^3 g$, so $\delta S/\delta g \sim \partial^4 g$, giving fourth-order equations. □

Lemma 7.7 (3D Curvature Identity)**.** *In three spacetime dimensions, the Weyl tensor vanishes identically and the Riemann tensor is completely determined by the Ricci tensor:*

$$R_{\mu\nu\rho\sigma} = g_{\mu\rho}R_{\nu\sigma} - g_{\mu\sigma}R_{\nu\rho} - g_{\nu\rho}R_{\mu\sigma} + g_{\nu\sigma}R_{\mu\rho} - \tfrac{R}{2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}).$$

Every diffeomorphism-invariant scalar therefore reduces to a function of R and $R_{\mu\nu}R^{\mu\nu}$.

Lemma 7.8 (Quadratic Curvature Terms)**.** *Quadratic curvature invariants such as $R_{\mu\nu}R^{\mu\nu}\sqrt{|g|}$ generate fourth-order field equations and are excluded by Axiom 4.*

Proof. $R_{\mu\nu}R^{\mu\nu} \sim (\partial^2 g)^2$, so variation with respect to $g^{\mu\nu}$ involves $\partial^4 g$ terms. □

7.3 Main Theorem: Uniqueness of the Interaction Action

Theorem 7.9 (Uniqueness of the Interaction Action Functional)**.** *Let $(\mathcal{M}, g)$ be the three-dimensional interaction manifold with coordinates (t, I_S^1, I_S^2) equipped with the block-diagonal metric of Section 4.3. Under Axioms 1–4, the unique admissible action functional is, up to a boundary term,*

$$S[g] = \int_{\mathcal{M}} \left(\frac{1}{2\kappa} R + \Lambda \right) \sqrt{|g|}\, \mathrm{d}^3 x, \tag{13}$$

where R is the Ricci scalar, κ is a coupling constant, and Λ is a cosmological-type constant. Variation of (13) *yields the* interaction field equation

$$R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \tag{14}$$

where $T_{\mu\nu}$ is the interaction source tensor (Definition 6.10).

Proof. The argument proceeds in seven steps.

Step 1 (Locality and Covariance). By Axioms 1–2, $\mathcal{L}$ must be built from the Riemann tensor and its covariant derivatives; Christoffel symbols are excluded by their non-tensorial transformation law.

Step 2 (Dimensional Reduction). By Lemma 7.7, every diffeomorphism-invariant scalar on a 3-manifold reduces to a function of R and $R_{\mu\nu}R^{\mu\nu}$.

Step 3 (Second-Order Constraint). By Lemma 7.6, all terms involving ∇R are excluded.

Step 4 (Quadratic Exclusion). Quadratic curvature terms are excluded by Lemma 6.5.

Step 5 (Admissible Lagrangians). The only surviving scalars linear in curvature (or zeroth order) are 1 (volume element contribution) and R, giving $\mathcal{L} = \alpha + \beta R$.

Step 6 (Normalisation). Setting $\beta = 1/(2\kappa)$ and $\alpha = \Lambda$ produces (13).

Step 7 (Variation). Standard variation of the Einstein–Hilbert term $\int R\sqrt{|g|}$ yields $\delta(\int R\sqrt{|g|})/\delta g^{\mu\nu} = (R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu})\sqrt{|g|}$; the cosmological term contributes $\Lambda g_{\mu\nu}$; the scalar-field action for φ produces $\kappa T_{\mu\nu}$ via (12). Uniqueness follows from Steps 1–6. □

Remark 7.10. Equation (14) is structurally identical to the Einstein field equations of general relativity, but it governs the geometry of the interaction space rather than spacetime. The coupling constant κ has units of curvature per unit interaction intensity squared.

The variation of the scalar-field action with respect to φ yields the *interaction wave equation*:

$$\Box_g \varphi := g^{\mu\nu}\nabla_\mu \nabla_\nu \varphi = 0. \tag{15}$$

Equations (14) and (15) together form a closed Einstein–scalar-field system [3].

8 Special Solutions

8.1 Flat Solution: Recovery of the Linear Framework

Proposition 8.1 (Flat-Space Limit)**.** *If $\varphi = \text{const}$, then $T_{\mu\nu} = 0$ and (14) reduces to $G_{\mu\nu} = 0$, i.e., $R_{\mu\nu} = 0$ (Ricci-flat). The manifold is locally isometric to $\mathbb{R}^n$, and the geodesic equation reduces to $\dot{I}_S = MI_S$ with constant M, recovering the original linear framework of [11].*

The entire prior three-paper series [10–12] is thus the zero-order approximation of the field theory, valid when emotional gradients are negligible and the interaction space is flat—the geometric analogue of the Newtonian limit of general relativity.

8.2 Ricci Flow and the Relaxation Solution

The Ricci flow [6, 8] is the geometric evolution equation

$$\frac{\partial g_{\mu\nu}}{\partial t} = -2R_{\mu\nu}. \tag{16}$$

It drives the metric toward flatness by smoothing regions of high curvature.

Theorem 8.2 (Relaxation as Ricci Flow)**.** *In the one-dimensional reduction, the unique solution $M(t) = (t + \tau_0)^{-1}$ of the relaxation condition $E(t) = 0$ corresponds to the Ricci flow on a conformally flat manifold. Specifically, setting $g_{11}(t) = e^{2u(t)}$, the Ricci flow gives*

$$\dot{u} = -K, \quad K(t) = \frac{1}{(t + \tau_0)^2},$$

whose solution $u(t) = \ln(t + \tau_0) + C$ implies $g_{11}(t) \propto (t + \tau_0)^2 \to \infty$ and $K \to 0$, i.e., the interaction space asymptotically flattens.

Proof. From $M = (t+\tau_0)^{-1}$ and $K = M^2$ (from the one-dimensional Ricci tensor $R_{11} = -\dot{K} - K^2 + K\,\mathrm{Tr}(K) = -(M^2)^{\cdot} - 0$ in 1D), the sectional curvature is $K = M^2 = (t+\tau_0)^{-2}$. Substituting into the Ricci flow equation $\partial_t g_{11} = -2R_{11} = 2Kg_{11}$ and writing $g_{11} = e^{2u}$ gives $\dot{u} = K = (t+\tau_0)^{-2}$. Integrating, $u(t) = -(t+\tau_0)^{-1} + C$, confirming $K \to 0$ as $t \to \infty$. $\square$

Remark 8.3. Theorem 8.2 provides a geometric derivation of the relaxation formula $M(t) = (t+\tau_0)^{-1}$ from first principles, as the Ricci flow trajectory on the one-dimensional conformally flat interaction manifold converging to the flat ($K = 0$) fixed point.

8.3 Constant Curvature Solutions

If $T_{\mu\nu} = \Lambda_{\text{eff}}\, g_{\mu\nu}$ for a constant Λ_{eff} (homogeneous interaction state), then (14) reduces to $R_{\mu\nu} = \Lambda_{\text{eff}} g_{\mu\nu}$, an Einstein space of constant Ricci curvature. In the context of interaction dynamics: $\Lambda_{\text{eff}} > 0$ gives positive curvature (convergent geodesics, stable coupled dynamics); $\Lambda_{\text{eff}} < 0$ gives negative curvature (divergent geodesics, escalating dynamics); $\Lambda_{\text{eff}} = 0$ recovers the flat case.

9 Discussion

9.1 Geometric Dictionary

Table 1 summarises the correspondence between geometric objects and their interaction-dynamics interpretations.

Geometric object	**Interaction interpretation**
Metric $G_{ij}(t, I_S)$	Structure of the interaction space
Flow tensor $K^i{}_j$	Rate of change of interaction coupling
Emotion tensor E_{ij}	Intrinsic interaction acceleration
R_{00}	Negative trace of total interaction acceleration
R_{ij}^{space}	Spatial interaction curvature
Source tensor $T_{\mu\nu}$	Gradient-driven interaction intensity
Cosmological constant Λ	Baseline interaction intensity
Geodesic completeness	Stable (non-singular) interaction trajectory

Table 1: Geometric dictionary for two-agent interaction dynamics.

9.2 The Dyadic Case: A Dimensional Subtlety

A critical dimensional subtlety arises for the two-agent ($n = 2$) dyad. When the interaction manifold is 2-dimensional (without the time direction), the Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ vanishes identically—a topological fact for any 2-dimensional Riemannian manifold. The field equation (14) therefore has no content in 2D.

For the dyadic case, the correct field equation on the two-dimensional interaction surface is the *Liouville equation*:

$$\Delta u = -\kappa T\, e^{2u}, \tag{17}$$

where $g = e^{2u}(\mathrm{d}x^2 + \mathrm{d}y^2)$ is the conformally flat representation of the metric, $K = -e^{-2u}\Delta u$ is the Gaussian curvature, and T is the scalar trace of the source. This equation has well-known explicit solutions for constant T (chronic coupling) and decaying $T \propto e^{-\alpha t}$ (relaxation dynamics).

The full three-dimensional framework of Section 7 applies without obstruction when the time coordinate is retained as part of the manifold, as in Stages I and II, since the manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{I_S}$ is three-dimensional. For $n \geq 3$ purely spatial agent dimensions, the Einstein equation holds without modification.

9.3 The Fisher–Weber–Fechner Metric

The metric g on the interaction manifold is not arbitrary: it is canonically determined by the information geometry of the perceptual encoding. Under Weber–Fechner encoding $I_{Si} = k_{S,i} \ln(S_i/S_0)$, the Fisher information metric on the stimulus space $\mathbb{R}^n_+$ is

$$g_{ij}(x) = \frac{k_{P,i}^2 \alpha_i^2}{k_{S,i}^2}\, \delta_{ij} = \gamma_{ii}^2\, \delta_{ij}. \tag{18}$$

In the isotropic case $k_{P,i}\alpha_i = k_{S,i}$, this reduces to the flat Euclidean metric, recovering Stage I. For heterogeneous agents, a curved manifold arises—for a two-agent dyad with heterogeneous sensitivities, the metric is a product of two copies of the Poincaré upper half-plane $\mathbf{H}^1$, giving constant negative sectional curvature $K = -1$, consistent with the conflict-prone nature of heterogeneous dyadic interaction [5].

9.4 Limitations

1. *Dimensionality.* The Einstein field equation is non-trivial only for manifold dimension $\dim \mathcal{M} \geq 3$. For the purely spatial dyadic case ($n = 2$), Equation (17) is the correct replacement, as discussed above.

2. *Signature.* The Riemannian (positive-definite) metric does not encode a preferred direction of time within the geometry. A Lorentzian signature $(-, +, \ldots, +)$ would allow causal propagation of interaction signals, at the cost of additional technical complexity and potential issues with geodesic completeness.

3. *Empirical accessibility.* The metric $g_{\mu\nu}$ and curvature $R_{\mu\nu\rho\sigma}$ are not directly measurable from behavioural data. Parameter estimation requires encoding observational data (physiological signals, linguistic affect scores) as geodesic curves and inferring g via multidimensional scaling or Riemannian manifold learning [2].

4. *Scalar field action.* The choice $S[\varphi] = \frac{1}{2}\int |\mathrm{d}\varphi|_g^2$ is the minimal (massless, no potential) scalar action. Richer models include a potential $V(\varphi)$ encoding nonlinear self-coupling, saturation, or threshold effects, leading to Klein–Gordon-type equations.

9.5 Future Directions

1. *Stochastic extension.* Replacing the geodesic equation with a stochastic differential equation on $(\mathcal{M}, g)$ (Eells–Elworthy–Malliavin framework) models interaction volatility as geodesic diffusion.

2. *Higher-dimensional agent networks.* For $n > 2$ purely spatial agent dimensions, the full Einstein system with non-trivial Weyl curvature governs group interaction dynamics, potentially encoding conformity, influence hierarchies, and social contagion.

3. *Empirical validation.* Dyadic physiological recordings (heart rate, skin conductance, facial EMG) can be used as state vectors; geodesic fitting via Riemannian manifold learning [2] provides a route to estimating g and κ from data.

4. *Matrix Ricci flow and optimal transport.* Connections to matrix Ricci flow [8, 9] and the Wasserstein geometry of probability measures on $\mathcal{M}$ may provide new tools for analysing transient interaction dynamics.

5. *Free energy principle.* The interaction manifold with Fisher metric coincides with the statistical manifold of Friston's active inference framework [4]. The field equations may thus provide a field-theoretic extension of free-energy minimisation for dyadic systems.

10 Conclusion

We have developed a unified geometric theory of two-agent interaction dynamics in three successive stages, culminating in an Einstein-type interaction field equation derived from first principles.

In Stage I, a time-dependent Riemannian metric is constructed from the linear interaction constraint $I_P = \gamma(t) I_S$; the interaction flow tensor $K^i{}_j$ and the emotion quantity tensor E_{ij} emerge as the natural geometric objects governing all Ricci curvature. In Stage II, the extension to $\gamma = \gamma(t, I_S)$ generates non-trivial spatial curvature; the Ricci tensor decomposes cleanly into temporal and spatial contributions. In Stage III, the interaction framework is placed in a coordinate-free Riemannian setting: the emotion tensor $E = \dot{M} + M^2$ is proved to equal the geodesic deviation operator (Theorem 6.5), identifying emotion as the curvature of interaction space.

Under four explicit axioms (locality, diffeomorphism covariance, metric dynamics, second-order field equations), the interaction geometry on the three-dimensional manifold $\mathcal{M}$ is uniquely governed by

$$R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu},$$

where $T_{\mu\nu}$ is the canonical stress-energy tensor of the interaction scalar field (12). All prior linear results are recovered as the flat-space limit, and the relaxation formula $M(t) = (t + \tau_0)^{-1}$ is derived as the asymptotic fixed point of Ricci flow.

The framework establishes two-agent interaction dynamics as a phenomenon governed by geometric field equations, placing it on the same mathematical footing as classical field theories while remaining grounded in the psychophysical structure of the prior hierarchy.

A Four-Layer Hierarchy Summary

Layer	Equation	Mathematical Object	Phenomenon
I: Discrete	$I_S(t+1) = \gamma I_S(t)$	Matrix semigroup	Turn-based interaction
II: Continuous	$\dot{I}_S = M(t) I_S$	ODE / semigroup generator	Real-time affective flow
III: Emotional	$\ddot{I}_S = E(t) I_S$	Second-order ODE	Emotional acceleration
IV: Geometric	$G_{\mu\nu} = \kappa T_{\mu\nu}$	Riemannian field equation	Curvature of interaction space

Table 2: Four-layer hierarchy of the psychophysical interaction theory.

B Correspondence Table

Linear / algebraic	Geometric / covariant
$I_S(t) \in \mathbb{R}^n$	$X(t) = \dot{\sigma}(t) \in T_{\sigma(t)}\mathcal{M}$
$I_P = \gamma I_S$	$\omega = g^\flat(X)$
Resistivity matrix γ	Metric tensor $g_{\mu\nu}$
Flow matrix $M(t)$	Connection coefficients $\Gamma^i{}_{jk}\dot{\sigma}^j$
$\dot{I}_S = M I_S$	Geodesic equation $DX/\mathrm{d}t = 0$
Emotion tensor $E = \dot{M} + M^2$	Geodesic deviation $\mathcal{E}^i{}_k = R^i{}_{jkl}\dot{\sigma}^j\dot{\sigma}^l$
$\ddot{I}_S = E I_S$	Jacobi equation $D^2 J/\mathrm{d}t^2 = R(\dot{\sigma}, J)\dot{\sigma}$
Conflict parameter $\mathcal{C}$	Sectional curvature $K < 0$
Spectral stability $\rho(\gamma) < 1$	Geodesic completeness
Relaxation $M = (t + \tau_0)^{-1}$	Ricci flow convergence to flat metric
Linear framework	Flat-space limit $R_{\mu\nu} = 0$

Table 3: Dictionary between the linear and geometric frameworks.

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